

ETL and Metadata in Data Warehousing

1. Slide: Introduction to ETL

Description:

ETL stands for **Extract, Transform, Load**. It is the backbone process in data warehousing to move data from multiple sources to a central repository (Data Warehouse).

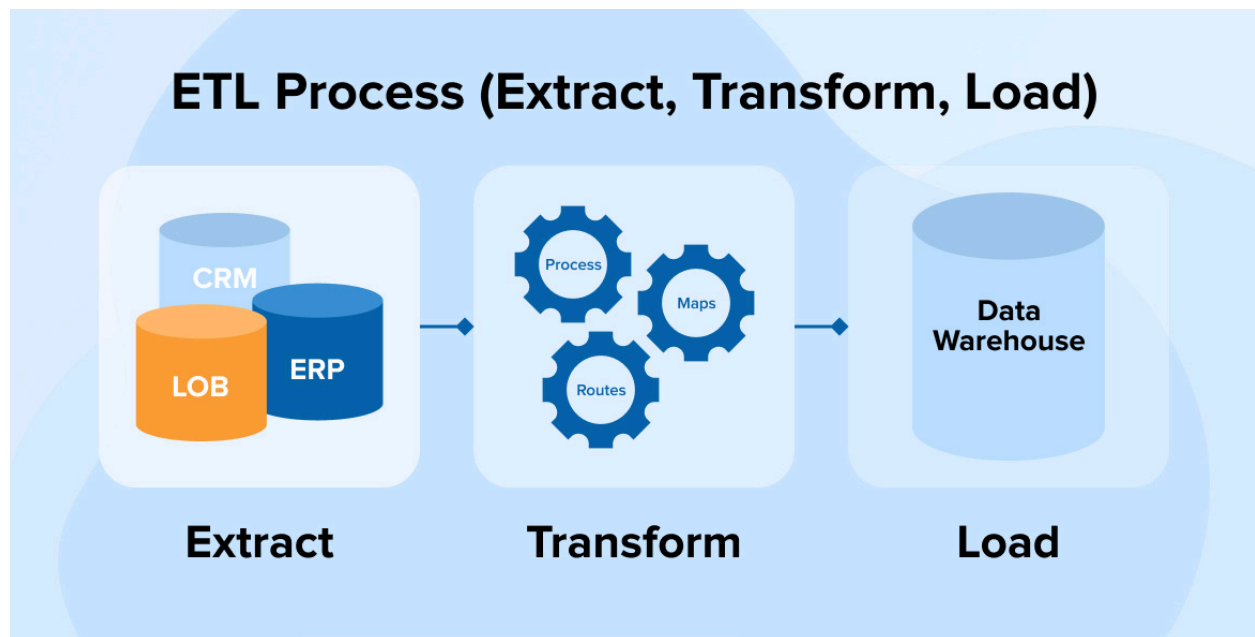


Illustration:

- Flowchart: Source Systems → Extract → Transform → Load → Data Warehouse

2. Slide: ETL Process Breakdown

Description:

Step	Purpose	Example
Extract	Pull data from source systems	Flat files, APIs, DBs
Transform	Clean, validate, format data	Removing NULLs, standardizing names

Load	Push processed data to target DW	Bulk INSERT, Merge, SCD handling
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Illustration:

Use gear icons for each step with a data pipeline in between.

Code Sample (Transformation):

```
-- Example transformation
SELECT
  UPPER(customer_name) AS customer_name,
  CASE WHEN country IS NULL THEN 'Unknown' ELSE country END AS country
FROM staging_customers;
```

3. Slide: Metadata in ETL

Description:

Metadata is "data about data." In ETL, metadata tracks how data is:

- Sourced
- Transformed
- Loaded
- Scheduled

Types in ETL:

- **Technical Metadata:** Data types, schema, lineage
- **Operational Metadata:** ETL job logs, load time, run status
- **Business Metadata:** Descriptions, business rules

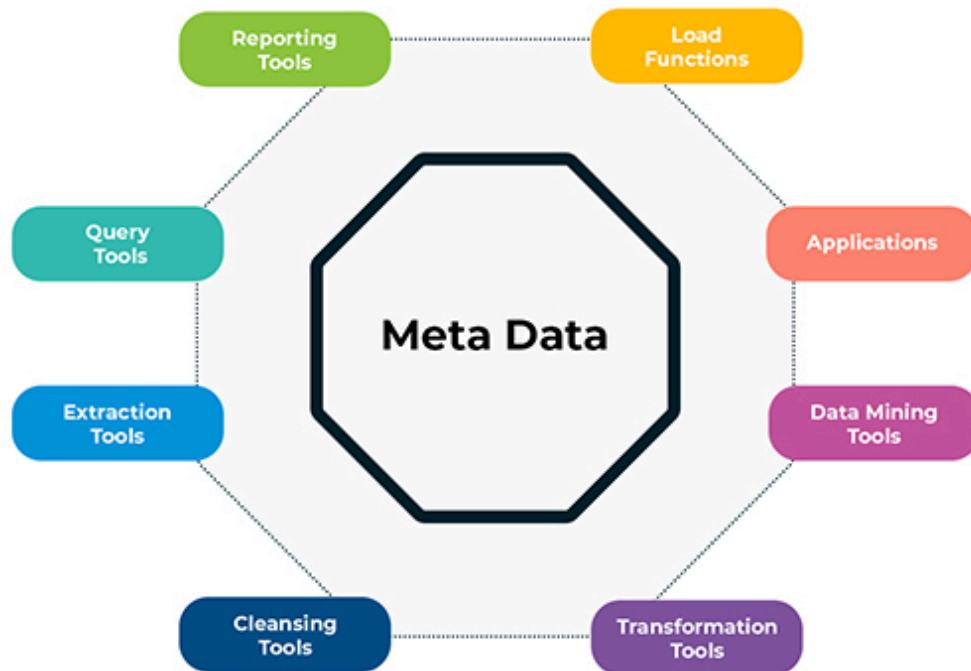


Illustration:

Metadata layers in ETL: Technical → Operational → Business

Sample Metadata Table:

```
CREATE TABLE etl_job_log (  
  job_id SERIAL PRIMARY KEY,  
  job_name VARCHAR(100),  
  start_time TIMESTAMP,  
  end_time TIMESTAMP,  
  status VARCHAR(20)  
);
```

4. Slide: Metadata in Data Warehousing

Description:

Metadata plays a critical role in:

- Maintaining data lineage
- Auditing
- Impact analysis
- Data governance

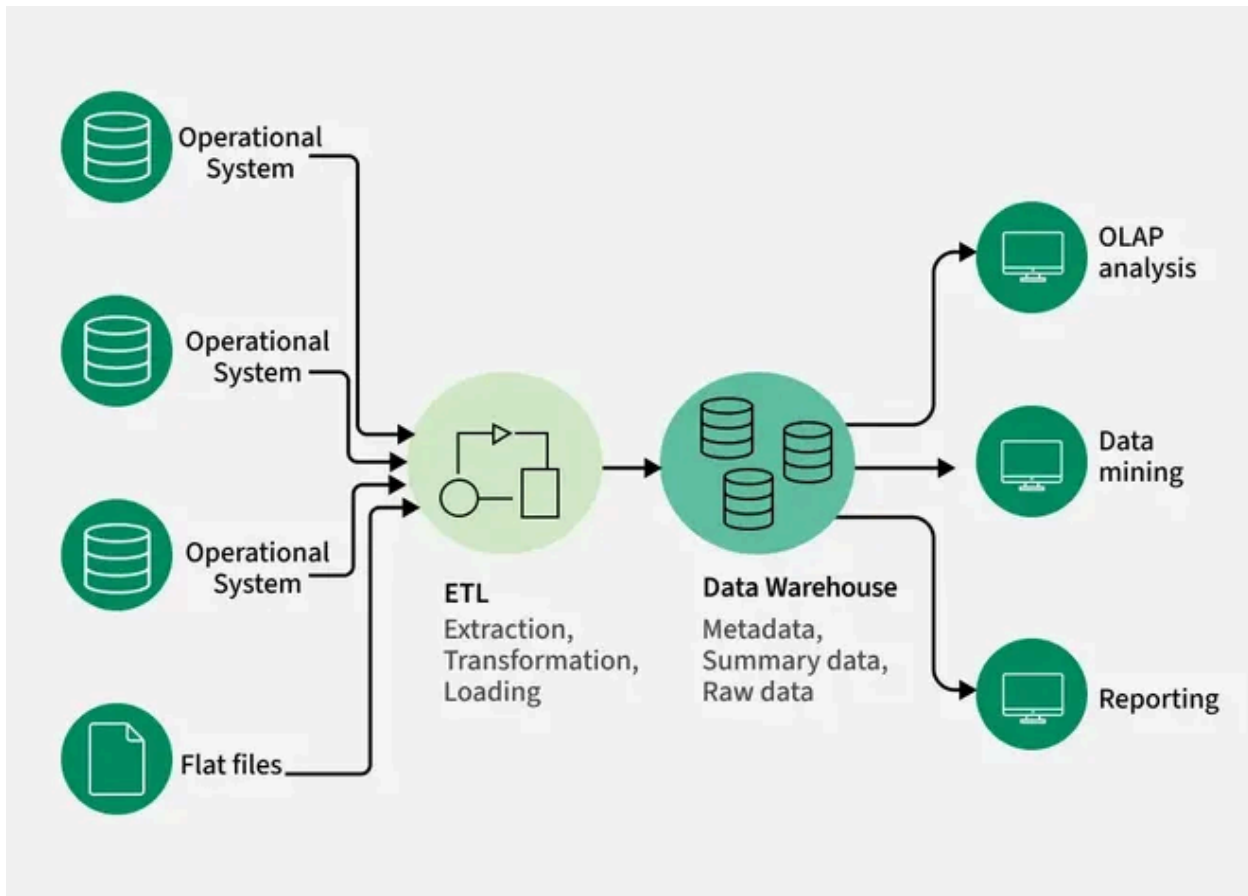


Illustration:

A diagram showing Data Sources → ETL → Metadata Repository → BI Tools

Code Sample (Query Lineage Audit):

```
-- Trace which job updated a record
SELECT job_name, end_time
FROM etl_job_log
WHERE end_time > (SELECT MAX(last_updated) FROM dim_customer);
```

5. Slide: Simple Data Warehouse Model

Description:

Basic star schema includes:

- Fact Table (e.g., `fact_sales`)
- Dimension Tables (`dim_customer`, `dim_product`, etc.)

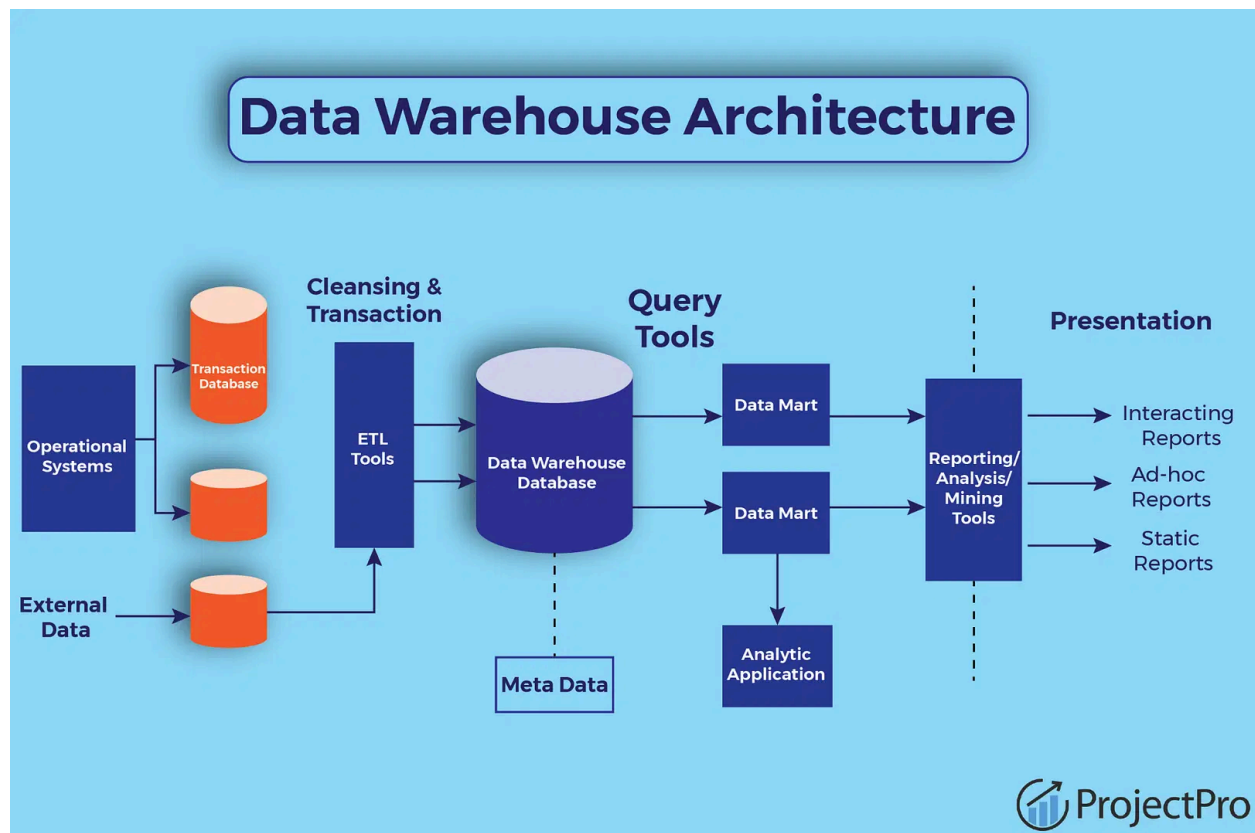


Illustration:

Star schema model with:

- `fact_sales`
- `dim_customer, dim_product, dim_time`

SQL Example:

```
-- Fact table
CREATE TABLE fact_sales (
  sale_id SERIAL,
  product_id INT,
  customer_id INT,
  sales_amount DECIMAL,
  sale_date DATE
);

-- Dimension
CREATE TABLE dim_customer (
  customer_id INT PRIMARY KEY,
```

```
customer_name TEXT,  
region TEXT  
);
```

6. Slide: Tools for ETL and Metadata Management

ETL Tools:

- Apache NiFi, Talend, Informatica, AWS Glue, Apache Airflow

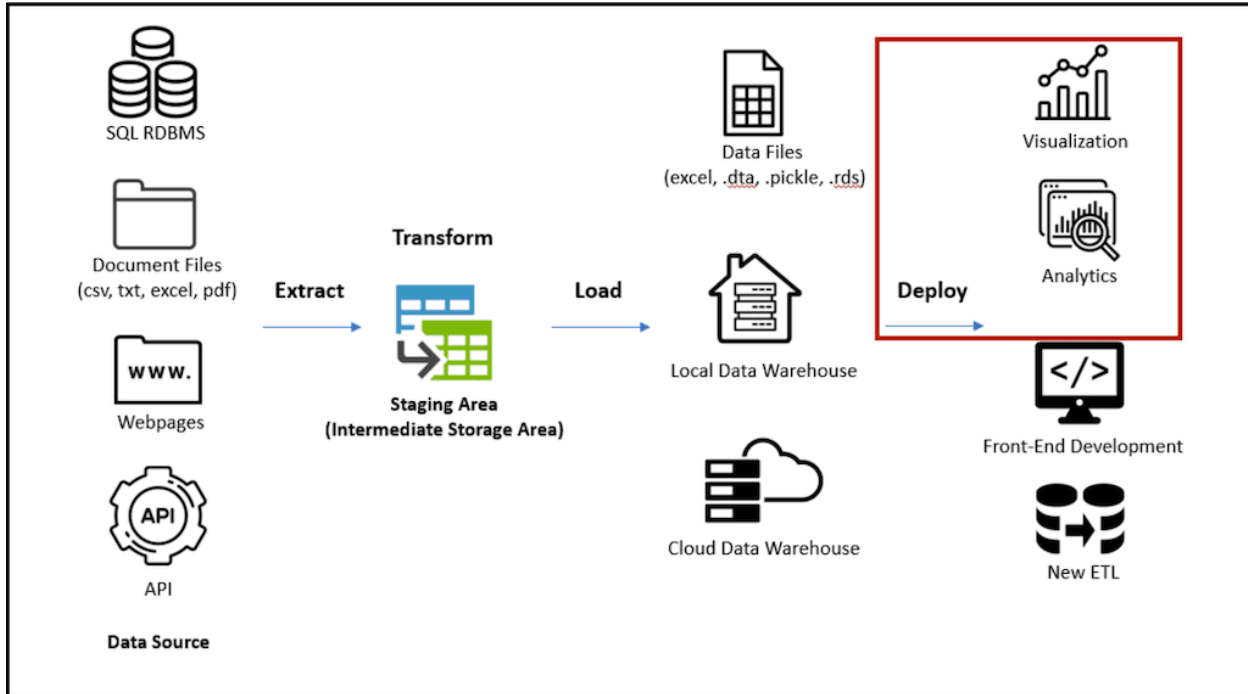
Metadata Management Tools:

- Apache Atlas
 - Alation
 - Collibra
 - Informatica EDC
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7. Slide: Real-World Scenario – Sales Analytics ETL

Scenario:

- Source: Online sales portal
- ETL: Extract order details, enrich with customer data, load to DW
- Metadata: Logs job run time, transformations, and business rules applied



dlllustration:

Scenario Diagram with Data Flow, Transformation Rules, Job Metadata Log

8. Slide: Summary

- ETL moves and transforms raw data into usable format
- Metadata tracks structure, meaning, and history of data
- Both are critical for building trustable, maintainable data warehouses

9. Slide: Optional Add-on – Excel Dashboard Sample

Include:

- Pivot table showing sales by region
 - Data load time and job status from metadata
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